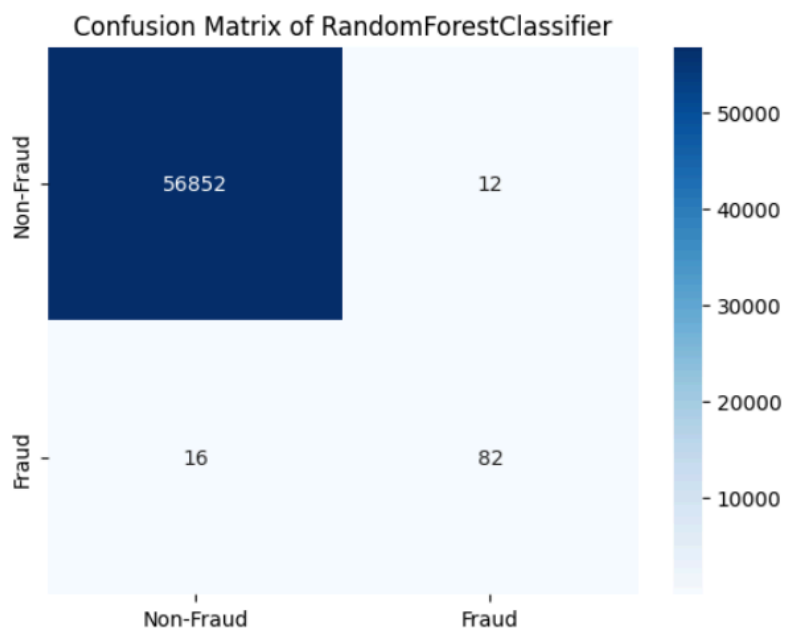
Model Evaluation

1. Random Forest

```
Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00     56864
           1       0.87      0.84      0.85        98

    accuracy                           1.00     56962
   macro avg       0.94      0.92      0.93     56962
weighted avg       1.00      1.00      1.00     56962
```




```
ROC-AUC Score of RandomForestClassifier:  0.9183
```

Performance: Balanced precision (87%) and recall (84%), F1-score 85%.

Business Impact: Strong fraud detection with very few false alarms - customers are protected without unnecessary disruptions.

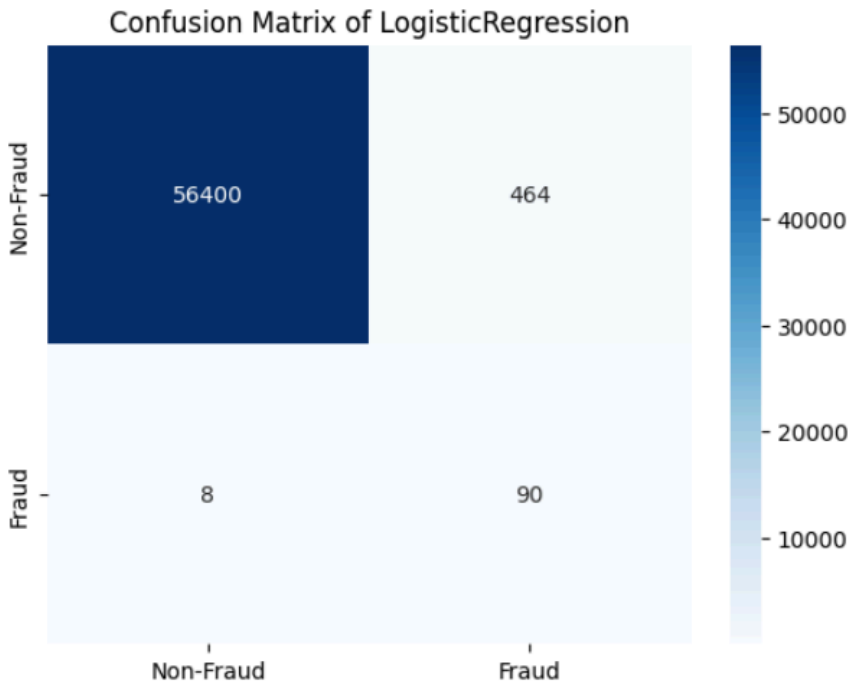
Takeaway: Reliable and regulator-friendly, ideal for deployment.

2. Logistic Regression

```
LogisticRegression Report:
              precision    recall  f1-score   support

           0       1.00      0.99      1.00     56864
           1       0.16      0.92      0.28        98

    accuracy                           0.99     56962
   macro avg       0.58      0.96      0.64     56962
weighted avg       1.00      0.99      0.99     56962
```




```
ROC-AUC of LogisticRegression: 0.95510376350878
```

Performance: Very high recall (92%) but poor precision (16%).

Business Impact: Catches nearly all fraud but generates too many false alarms - customers may lose trust.

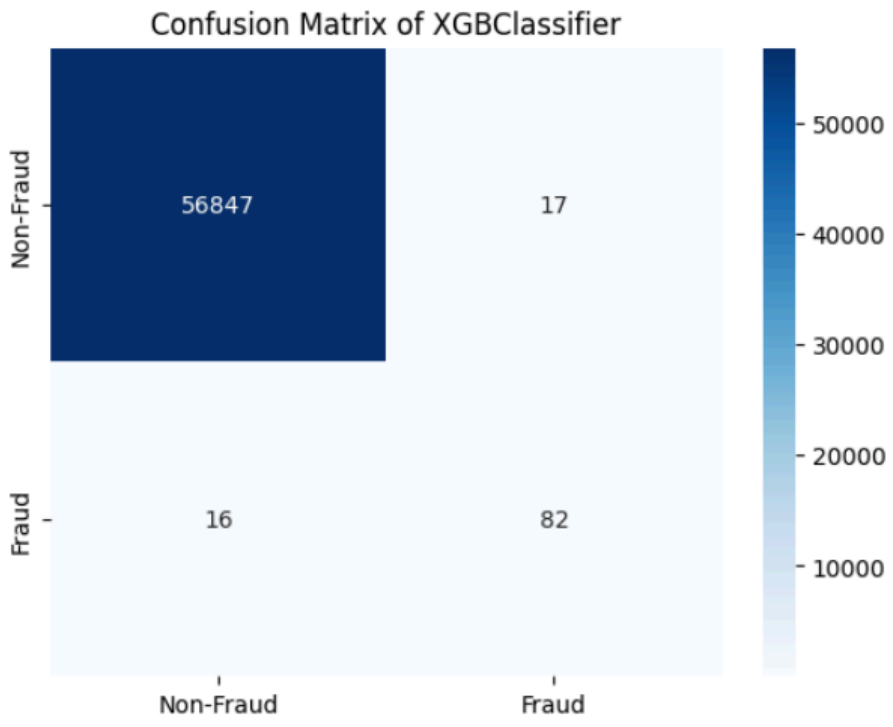
Takeaway: Useful as a baseline model, but not practical for production.

3. XGBoost

```
XGBoost Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00     56864
           1       0.83      0.84      0.83        98

    accuracy                           1.00     56962
   macro avg       0.91      0.92      0.92     56962
weighted avg       1.00      1.00      1.00     56962
```




```
ROC-AUC of XGBClassifier: 0.9182178674790119
```

Performance: Precision (83%), recall (84%), F1-score 83%**.**

Business Impact: Efficient and widely used in fintech, strong detection with few errors.

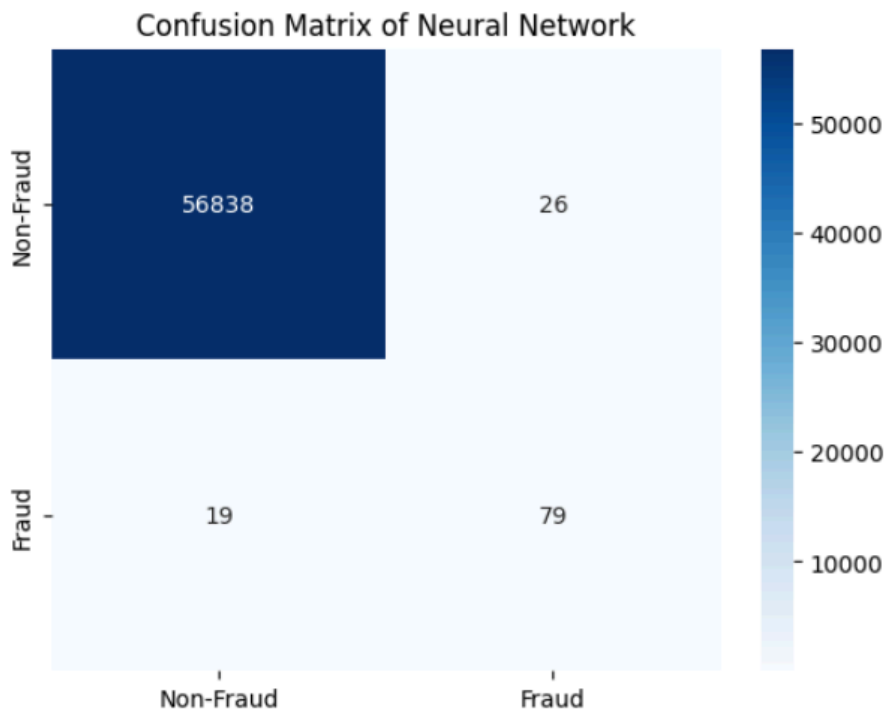
Takeaway: Competitive with Random Forest, but slightly less interpretable for compliance teams.

4. Neural Network

```
Neural Network Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00     56864
           1       0.75      0.81      0.78        98

    accuracy                           1.00     56962
   macro avg       0.88      0.90      0.89     56962
weighted avg       1.00      1.00      1.00     56962
```



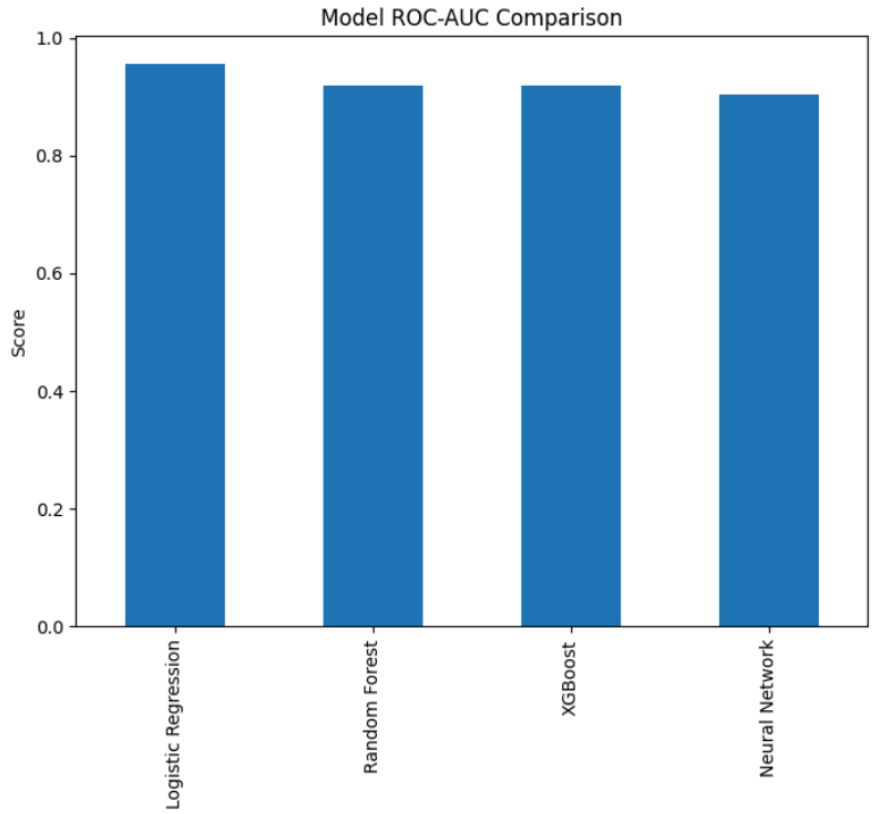

```
ROC-AUC of Neural Network: 0.9028326088454516
```

Performance: Precision (75%), recall (81%), F1-score 78%, lowest ROC-AUC.

Business Impact: Decent detection but weaker than tree-based models, harder to explain to regulators.

Takeaway: Adds complexity without clear business advantage.

Comparison



Logistic Regression vs Random Forest : Logistic Regression catches more fraud but frustrates customers with false alarms. Random Forest balances detection and convenience.

Random Forest vs XGBoost : Both strong, but Random Forest is more interpretable and slightly better suited for compliance.

Tree-Based Models vs Neural Network : Neural Network is less effective and harder to justify to regulators.

Overall Conclusion

Fraud detection is a **rare-event**, **business-critical challenge.**

Logistic Regression shows why recall alone is not enough.

Random Forest and XGBoost deliver balanced, reliable performance.

Neural Networks add complexity without clear gains.

Final Recommendation:

Deploy **Random Fores**t as the primary fraud detection model. It ensures customer convenience (low false alarms) and risk management (high fraud detection), while being transparent and regulator-approved.